Q.1 Explain Observables.

In quantum mechanics, an observable is any physical quantity that can be measured, such as: Position, Momentum, Energy, Spin of an electron, Polarization of a photon. The most common observable is the state of a qubit (measured as 0 or 1 in the computational basis) in quantum computing.

Mathematically, an observable is represented by a **Hermitian operator** acting on quantum states. The measurement of an observable in a quantum system corresponds to projecting the quantum state onto the eigenstates of the operator and obtaining one of its eigenvalues as the result of the measurement.

OR The measurement of an observable corresponds to projecting the quantum state onto its eigenstates. The measurement outcome is one of the eigenvalues of the observable.

Mathematical Representation of an Observable:

~~In quantum mechanics, observables are represented by **Hermitian operators**.~~ If $\hat{A}$ is an observable, it must satisfy the condition:

$$\hat{A} = \hat{A}^{\dagger}$$

where $\hat{A}^{\dagger}$ is the **adjoint** (complex conjugate transpose) of the operator $\hat{A}$.

The **eigenvalues** of the operator correspond to the **possible measurement outcomes**.

The **eigenvectors** correspond to the **quantum states** in which the system will collapse after measurement.

3. Example with Qubits

For a single qubit, an important observable is the **Pauli-Z operator**:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- Eigenvalues: $+1, -1$
- Eigenvectors: $|0\rangle$ and $|1\rangle$

👉 Meaning: measuring a qubit in the Z-basis gives either **0 or 1**, with probabilities determined by the qubit's state.

Q.2 Explain Pauli operators. / Q.3 Explain Pauli matrices in detail

The **Pauli operators** are a set of fundamental **2×2 Hermitian matrices** in **quantum computing** that act on **qubits**. They play a crucial role in the manipulation of qubits and are used extensively in quantum gates, quantum circuits and quantum algorithms. The Pauli operators are the building blocks of quantum operations with Pauli matrices referring to the explicit matrix representations of the fundamental Pauli operators.

Pauli Operators:

1. Pauli-X (Not Gate):

   The **Pauli-X** operator is often referred to as the **NOT gate** because it flips the state of a qubit between the $|0\rangle$ and $|1\rangle$ states.

The matrix representation of the Pauli-X operator is:

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Action on quantum states:

- $\hat{X}|0\rangle = |1\rangle$
- $\hat{X}|1\rangle = |0\rangle$

2. **Pauli-Y:**

   The **Pauli-Y** operator is associated with a rotation in the complex plane and is often used in quantum algorithms to **phase flip but also introduces a phase factor i**

The matrix representation of the Pauli-Y operator is:

$$\hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Action on quantum states:

- $\hat{Y}|0\rangle = i|1\rangle$
- $\hat{Y}|1\rangle = -i|0\rangle$

3. **Pauli-Z (Phase Flip):**

   The **Pauli-Z** operator is used to perform a **phase flip** on the quantum state, meaning it does not affect the probability of measuring the qubit in the computational basis, but it does introduce a relative phase between the states.

The matrix representation of the Pauli-Z operator is:

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Action on quantum states:

- $\hat{Z}|0\rangle = |0\rangle$
- $\hat{Z}|1\rangle = -|1\rangle$

4. Additionally, the **identity matrix** is included:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Together $\{I, \sigma_x, \sigma_y, \sigma_z\}$ form the Pauli set .

Importance in Quantum Computing

- **Quantum gates:** The Pauli operators form the building blocks of more complex gates.
- **Error correction:** Errors in quantum systems are often modeled as Pauli errors (X, Y, Z).

- **Measurements:** Measuring a qubit in the computational basis corresponds to the Pauli-Z operator.
- **Spin systems:** They describe electron spin and are central in physics as well as computing.

3. Properties of Pauli Matrices

1. **Hermitian** ($\sigma_i = \sigma_i^\dagger$)

   → Can represent **observables** (measurable quantities).
2. **Unitary** ($\sigma_i^\dagger \sigma_i = I$)

   → Can represent **quantum gates** (reversible operations).
3. **Square to Identity:**

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I$$

4. **Determinant:**

   $\det(\sigma_i) = -1.$
5. **Traceless:**

   $\mathrm{Tr}(\sigma_i) = 0.$
6. **Non-commutative:**

   Pauli matrices do not commute. Instead, they satisfy the **commutation relations**:

- Example:

$$[\sigma_x, \sigma_y] = 2i\sigma_z$$

- This reflects the uncertainty principle (cannot measure spin in two directions simultaneously).

4. **Geometric Meaning:**

   On the **Bloch sphere**, each Pauli operator corresponds to a **π rotation (180°) around an axis**:
   - σ_x → rotation about the x-axis
   - σ_y → rotation about the y-axis
   - σ_z → rotation about the z-axis

Q.4 Explain outer products and matrix representations.

Outer products are an essential mathematical tool used to describe quantum states, operators, and their interactions. The outer product, along with matrix representations, is crucial for understanding how quantum operations are performed.

Outer products

The outer product (or dyadic product) is an operation between two vectors that results in a matrix. It is fundamentally different from the dot product, which produces a scalar.

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \langle\phi| = \begin{pmatrix} \gamma^* & \delta^* \end{pmatrix}$$

$$|\psi\rangle\langle\phi| = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \begin{pmatrix} \gamma^* & \delta^* \end{pmatrix} = \begin{pmatrix} \alpha\gamma^* & \alpha\delta^* \\ \beta\gamma^* & \beta\delta^* \end{pmatrix}$$ **(OR)**

Definition:
If $|a\rangle$ and $|b\rangle$ are two **column vectors**, their **outer product** is denoted as $|a\rangle\langle b|$, and it results in a **matrix**.

The outer product is mathematically defined as:

$$|a\rangle\langle b| = \text{matrix formed by multiplying the column vector } |a\rangle \text{ by the conjugate transpose of } |b\rangle.$$

- If $|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$ and $|b\rangle = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$, then the outer product $|a\rangle\langle b|$ results in an $n \times n$ matrix:

$$|a\rangle\langle b| = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \begin{pmatrix} b_1^* & b_2^* & \cdots & b_n^* \end{pmatrix} = \begin{pmatrix} a_1 b_1^* & a_1 b_2^* & \cdots & a_1 b_n^* \\ a_2 b_1^* & a_2 b_2^* & \cdots & a_2 b_n^* \\ \vdots & \vdots & \ddots & \vdots \\ a_n b_1^* & a_n b_2^* & \cdots & a_n b_n^* \end{pmatrix}$$

Example:
Let's consider two qubits, $|0\rangle$ and $|1\rangle$, which are represented by the following vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The **outer product** of $|0\rangle$ and $|1\rangle$ is:

$$|0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

2. Matrix Representations

Definition

In quantum mechanics, every **linear operator** can be represented as a **matrix** relative to a chosen basis.

- If the basis is $\{|0\rangle, |1\rangle\}$, then any operator $\hat{A}$ can be written as a 2×2 matrix:

$$A = \begin{pmatrix} \langle 0|\hat{A}|0\rangle & \langle 0|\hat{A}|1\rangle \\ \langle 1|\hat{A}|0\rangle & \langle 1|\hat{A}|1\rangle \end{pmatrix}$$

Examples

1. **Pauli-X operator**

   Acts as a NOT gate:

$$\sigma_x|0\rangle = |1\rangle, \quad \sigma_x|1\rangle = |0\rangle$$

   Matrix form:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Q.6 Define a Hilbert space. What are its essential properties?

A **Hilbert space** is a special type of **vector space** equipped with an **inner product**, which allows us to measure norms and orthogonality between vectors. It is a **complete inner product space** over the field of complex numbers C. The Hilbert space is **complete**, meaning that it contains all limits of Cauchy sequences of vectors.

In quantum computing, The state of a quantum system (like a qubit) is represented by a **vector in a Hilbert space**. For a single qubit, the Hilbert space is **2-dimensional** (spanned by $|0\rangle$ and $|1\rangle$).For n qubits, the Hilbert space has dimension 2^n.

Essential Properties of a Hilbert Space

1. **Vector Space Properties :Closure(** Adding two vectors or multiplying by a scalar gives another vector in the space), **Associativity & Commutativity of addition**, **Existence of zero vector** and **additive inverse** and **Distributivity** of scalar and vector addition.
2. **Inner Product**:

   A Hilbert space H is equipped with an **inner product** that maps two vectors $|\psi\rangle$and $|\phi\rangle$ to a complex number $\langle\psi|\phi\rangle$.

3. **Norm**:

   The **norm** of a vector $|\psi\rangle$|\psi\rangle$|\psi\rangle$ is defined as:

   $$\|\psi\| = \sqrt{\langle\psi \mid \psi\rangle}$$

   This defines the **length** of the vector in the Hilbert space. The norm must satisfy **Positivity**

4. **Orthogonality**:

   Two vectors $|\psi\rangle$ and $|\phi\rangle$ in a Hilbert space are said to be **orthogonal** if their inner product is zero:

   $$\langle\psi|\phi\rangle=0$$

5. **Completeness**:

   A Hilbert space is **complete** in the sense that any Cauchy sequence has a limit that also belongs to the Hilbert space.

Example in Quantum Computing

Single qubit Hilbert space ($\mathbb{C}^2$):

Basis vectors:

$$|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}$$

Any state is:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1$$

Two qubits ($\mathbb{C}^4$):

Basis: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

Q.7 Explain hermitian, unitary, and normal operators.

In quantum computing operators are used to represent physical observables, quantum states, and quantum operations. Operators can be classified into different categories based on their properties.

1. Hermitian Operators

Definition:

A **Hermitian operator** (also called a **self-adjoint operator**) is an operator A that is equal to its own conjugate transpose. In simpler way we can say that A *Hermitian operator* is a special kind of operator that behaves like a mirror. When you have a quantum state, you can apply this operator to it, and in return it'll give you a new state.

$$A = A^\dagger$$

where $A^\dagger$ means transpose + complex conjugate.

Properties:

1. **Real Eigenvalues**: Hermitian operators have **real eigenvalues**
2. **Orthogonal Eigenvectors**: The eigenvectors corresponding to distinct eigenvalues of a Hermitian operator are **orthogonal** to each other.
3. Represent **observables** (measurable physical quantities).

Example

Pauli-Z operator:

$$\sigma_z = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}, \quad \sigma_z^\dagger = \sigma_z$$

In Quantum Computing

Measuring a qubit (e.g., in the Z-basis) is described by a Hermitian operator.

2. Unitary Operators

Definition

Which simply mean that if you take a unitary operator and multiply it by its adjoint, you get an identity. This means that you can always return to the original quantum state.

An operator (matrix) U is **Unitary** if its conjugate transpose is also its inverse:

$$U^\dagger U = UU^\dagger = I$$

Properties

- **Preserve inner products:**
  $\langle U\psi | U\phi \rangle = \langle \psi | \phi \rangle$.
- **Preserve length (norm):**
  $\| U|\psi\rangle \| = \| |\psi\rangle \|$.
- Represent **reversible quantum evolutions**.

Example

Hadamard gate:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H^\dagger H = I$$

In Quantum Computing

All **quantum gates** are unitary, ensuring no informat s lost during computation.

3. Normal Operators

Definition

An operator N is **Normal** if it **commutes with its conjugate transpose**:

$$NN^\dagger = N^\dagger N$$

Properties

- Includes both **Hermitian** and **Unitary** operators as special cases.
- Can be diagonalized using a unitary transformation.
- Eigenvectors form an **orthonormal basis**.

Example

Any diagonal matrix like:

$$N = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

is normal.

A normal operator is one that can be both *Hermitian* and *unitary* at the same time.

Q.9 Explain the trace of an operator. / Q.11 Explain the properties of the trace.

In quantum mechanics and quantum computing, the **trace** of an operator is a fundamental concept used in many calculations, including quantum measurements, density matrices, and expectation values.

The **trace** of a square matrix (or linear operator) A, denoted by Tr(A), is the **sum of its diagonal elements** in any orthonormal basis.

(OR.. Pehle General phir quantum)

For a square matrix (operator) A, the **trace** is defined as the sum of its diagonal elements:

$$\mathrm{Tr}(A) = \sum_i A_{ii}$$

If A is an operator on a vector space with basis $\{|i\rangle\}$, then:

$$\mathrm{Tr}(A) = \sum_i \langle i|A|i\rangle$$

Here:

- $\{|i\rangle\}$ is an **orthonormal basis** of the Hilbert space.
- $\langle i|\hat{A}|i\rangle$ is the **expectation value** of $\hat{A}$ in the state $|i\rangle$.

Properties of the Trace

1. Linearity

The trace is a **linear** operation, which means that it satisfies the following property:

$$\mathrm{Tr}(aA + bB) = a\,\mathrm{Tr}(A) + b\,\mathrm{Tr}(B)$$

The trace distributes over addition and scalar multiplication which is useful when working with sums of operators

Example:

$$\mathrm{Tr}(2I + 3\sigma_z) = 2\mathrm{Tr}(I) + 3\mathrm{Tr}(\sigma_z) = 2(2) + 3(0) = 4.$$

2. Cyclic Property

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This property allows for the rearrangement of operators inside a trace without changing the result. Even if AB≠ BA, their traces are equal

The trace has a **cyclic property**, meaning that for any square matrices $\hat{A}$ and $\hat{B}$, we have:

$$\mathrm{Tr}(\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{A})$$

This property extends to products of more than two operators as well:

$$\mathrm{Tr}(\hat{A}\hat{B}\hat{C}) = \mathrm{Tr}(\hat{B}\hat{C}\hat{A}) = \mathrm{Tr}(\hat{C}\hat{A}\hat{B})$$

3. Invariant under Basis Change:

For any unitary operator U:

$$\mathrm{Tr}(U^\dagger \hat{A} U) = \mathrm{Tr}(\hat{A})$$

The trace is an **intrinsic property** of the operator. This means trace is **independent of the basis** — it always gives the same result regardless of which orthonormal basis you choose.

4. **Relation to Eigenvalues**

The trace of a matrix equals the **sum of its eigenvalues** (counting multiplicities).

$$\mathrm{Tr}(A)=\lambda_1+\lambda_2+\cdots+\lambda_n$$

This is very important in quantum physics, since eigenvalues represent measurable quantities.

Traces of Special Operators

Identity matrix I: $\mathrm{Tr}(I) = n$ (dimension of space).

Pauli matrices: All have trace zero.

Density operator ρ: $\mathrm{Tr}(\rho) = 1$ (probabilities must sum to 1).

Q.13

In quantum computing, **commutators** are mathematical objects that describe how operators interact with each other. The **commutator** of two operators A and B is defined as:

$$[A, B] = AB - BA$$

Where, [A,B]is the **commutator**, A and B are operators, AB and BA are the products of the operators in different orders.

If [A,B]=0, we say that A and B **commute**. And if [A,B]≠0[A, B], they **do not commute**.

In quantum mechanics, non-commuting operators mean the corresponding physical observables cannot be measured precisely at the same time.

Properties of Commutators

1. Antisymmetry

The commutator is **antisymmetric**, meaning:

$[A, B] = -[B, A]$

This property tells us that the order in which you apply the operators matters: switching the order of operations changes the sign of the commutator.

2. **Commutator of a Product**

The commutator follows a **product rule** similar to the derivative:

$[AB,C]=A[B,C]+[A,C]B$

This property is similar to the Leibniz rule in calculus, where the commutator of a product of operators is split into two terms.

3. Commutator with Scalar

If λ is a scalar (a constant), the commutator between λ and any operator Ais zero:

$$[A, \lambda]=0$$

This is because scalar quantities do not affect the result of an operator's action.

4.

Commutators of the Pauli Matrices

The Pauli matrices satisfy special commutation relations:

$$[\sigma_x, \sigma_y] = 2i\sigma_z, \quad [\sigma_y, \sigma_z] = 2i\sigma_x, \quad [\sigma_z, \sigma_x] = 2i\sigma_y$$

These relations show how the Pauli operators interact and form the basic algebraic structure underlying qubit operations.